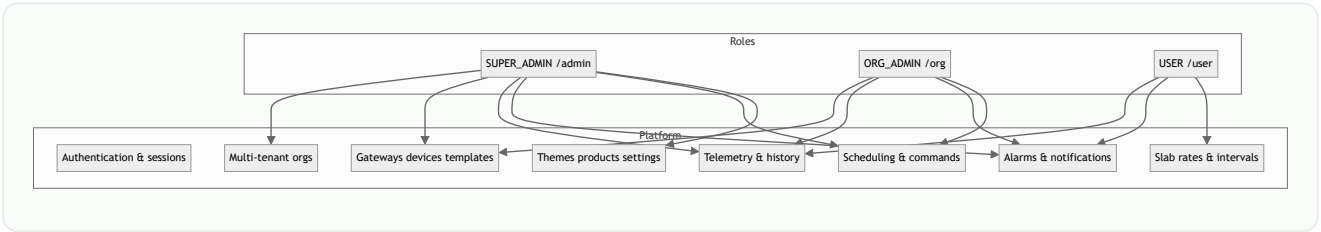


Application functionality

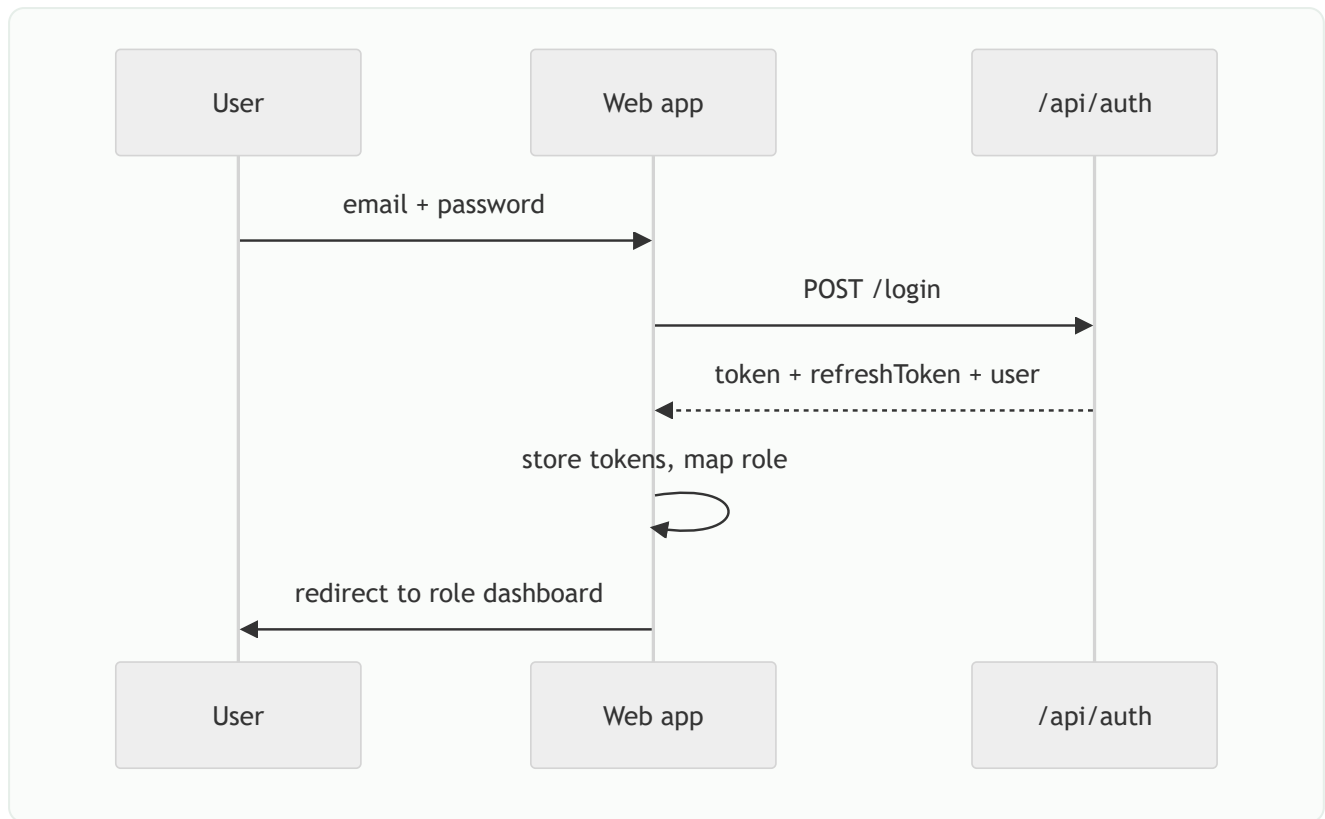
Detailed feature catalog for Smart AgriTech EMS, organized by role and module.

Feature map



Authentication & account

Feature	Who	Description
Login / logout	All	Email + password; JWT access + refresh
Session refresh	All	Silent token refresh on 401
Forgot / reset password	All	Email reset flow
Change password	All	<code>/user/account</code> and API
Role-based redirect	All	<code>/</code> → <code>/admin</code> , <code>/org</code> , or <code>/user</code>



Super Admin (`/admin`) — 22 nav items

Dashboard & monitoring

Page	Route	Features
Dashboard	<code>/admin</code>	Org/user/device/gateway counts; online/offline; power & variable charts for selected device; recent alarms; device switch toggles
Dashboard detail	<code>/admin/dashboard-detail</code>	Live variable cards from selected device; summary KPIs; time range
Data center	<code>/admin/data-center</code>	Cross-org device overview, latest readings
Sensor history	<code>/admin/sensor-history</code>	Paginated raw ingest log; CSV export
Historical data	<code>/admin/historical-data</code>	Per-device variable chart + table; date range

Organization & access

Page	Route	Features
Organizations	<code>/admin/organizations</code>	CRUD tenants

Page	Route	Features
Users	<code>/admin/users</code>	CRUD all roles; org assignment; status; password reset

IoT inventory

Page	Route	Features
Gateways	<code>/admin/gateways</code>	CRUD; serial; org binding
Devices	<code>/admin/devices</code>	CRUD; template; gateway; MQTT config modal; ingest key on create
Device detail	<code>/admin/devices/:id</code>	Overview, metrics, schedule, user assignment, remote switch
Device templates	<code>/admin/device-templates</code>	CRUD; clone
Template detail	<code>/admin/device-templates/:id</code>	Slaves & variables CRUD
Device timestamps	<code>/admin/device-timestamps</code>	Last-seen / connectivity log

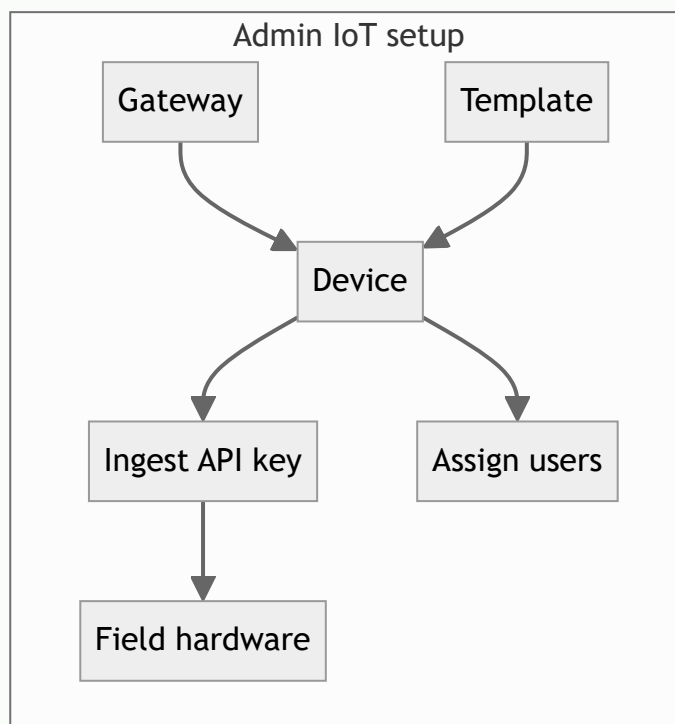
Alarms

Page	Route	Features
Variable alarms	<code>/admin/variable-alarms</code>	Active alarm list; process
Alarm history	<code>/admin/alarm-history</code>	Variable + linkage tabs; device filter
Linkage records	<code>/admin/linkage-records</code>	Automation linkage history
Template triggers	<code>/admin/template-triggers</code>	Threshold rules on template variables
Alarm settings	<code>/admin/alarm-settings</code>	Org alarm configuration
Alarm contacts	<code>/admin/alarm-contacts</code>	Notification contacts

Platform

Page	Route	Features
Icons	<code>/admin/icons</code>	Upload/manage UI icons (Cloudinary)
Products	<code>/admin/products</code>	Subscription product catalog
Schedule tasks	<code>/admin/schedule-tasks</code>	CRUD cron jobs; toggle; execution logs
Theme settings	<code>/admin/theme-settings</code>	Themes; assign to orgs

Page	Route	Features
Settings	<code>/admin/settings</code>	System key/value settings



Org Admin (`/org`) — 17 nav items

Same core IoT and alarm features as admin, **scoped to own organization**:

Module	Pages	Notes
Dashboard	<code>/org</code> , <code>/org/dashboard-detail</code>	Org device stats; monthly energy
Team	<code>/org/users</code>	Create <code>USER</code> role only
Widgets	<code>/org/widget-templates</code>	Dashboard widget definitions
Inventory	devices, gateways, templates	No cross-org access
History	sensor-history, historical-data, device-timestamps	Device-scoped charts
Alarms	template-triggers, settings, contacts, alarm-history	Org-scoped
Tasks	<code>/org/schedule-tasks</code>	Scheduled switch jobs
Settings	<code>/org/settings</code>	Org profile & preferences

End User (/user) — 18 nav items

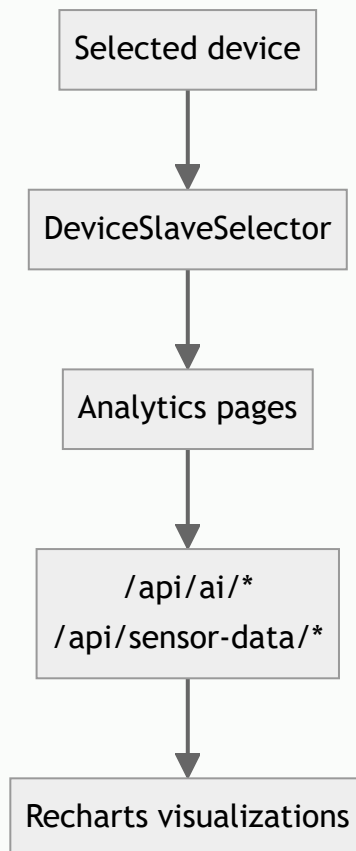
Focused on **assigned devices** and consumption analytics:

Dashboard & account

Page	Route	Features
Dashboard	/user	Greeting, subscription tier, stats, notifications preview, live charts
Dashboard detail	/user/dashboard-detail	Device variable grid
Account	/user/account	Profile, password
Notifications	/user/notifications	Mark read, list

Analytics (AI-style pages)

Page	Route	Data source
AI analytics	/user/ai-analytics	Dashboard summary + energy + PF + predictions; chat UI
Energy consumption	/user/energy-consumption	/api/ai/energy-consumption
Power factor	/user/power-factor	Gauge + trend + events
Voltage imbalance	/user/voltage-imbalance	Phase charts
Current imbalance	/user/current-imbalance	Current analytics
Anomalies	/user/anomalies	Acknowledge anomaly records



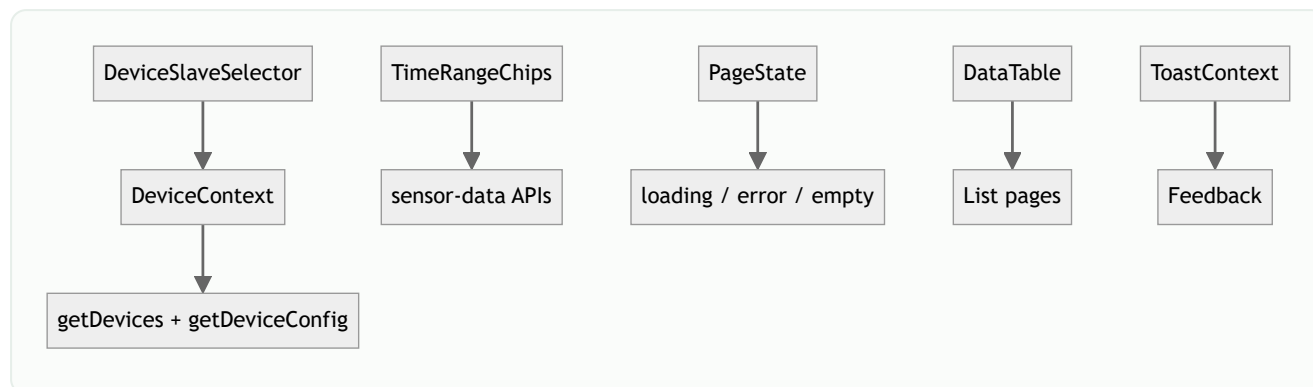
Billing & scheduling

Page	Route	Features
Slab rates	<code>/user/slab-rates</code>	Electricity tariff slabs CRUD
Interval history	<code>/user/interval-history</code>	Billing intervals + charts
Schedule	<code>/user/schedule</code>	View scheduled tasks
Subscription	<code>/user/subscription</code>	Plan status & history
Products	<code>/user/products</code>	Browse available plans

Alarms & history

Page	Route	Features
Sensor history	<code>/user/sensor-history</code>	Raw reading log
Alarm template	<code>/user/alarm-template</code>	View/toggle templates
Alarm settings	<code>/user/alarm-settings</code>	Reuses org alarm settings UI
Alarm history	<code>/user/alarm-history</code>	Device-filtered history

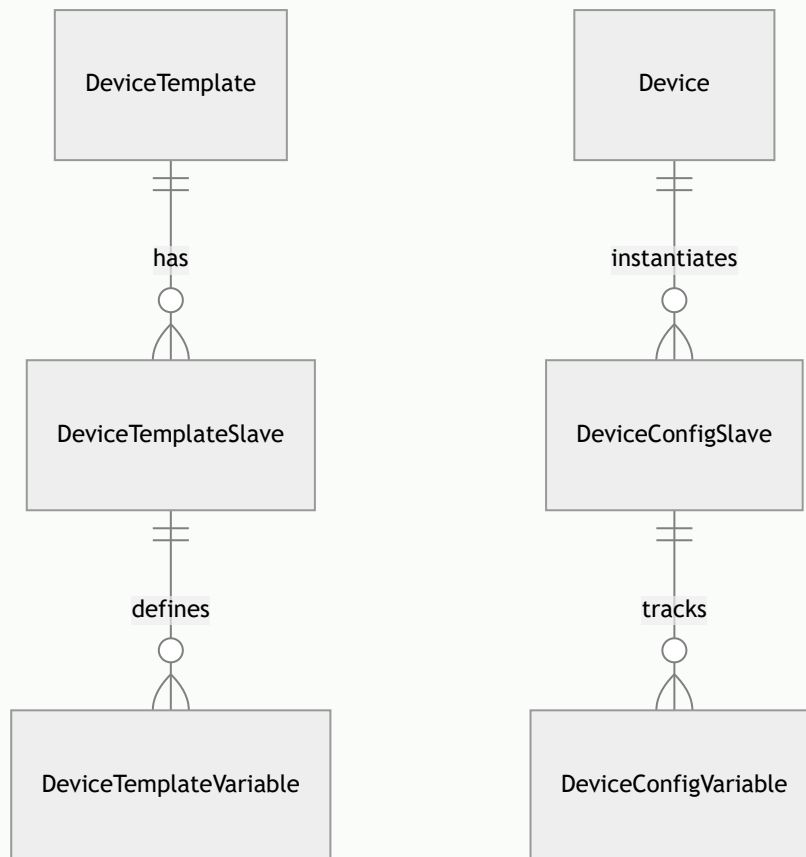
Shared components (all roles)



Component	Purpose
DeviceSlaveSelector	Global device + slave/meter picker
DeviceContext	Selected device state across pages
TimeRangeChips	24h / 7d / 30d filters
SocketBridge	Auto-connect socket on login
DashboardLayout	Sidebar from <code>navConfig.jsx</code>

Device template model (functionality)

Templates define **what variables** a device type exposes:



Example — **SMM soil sensor:**

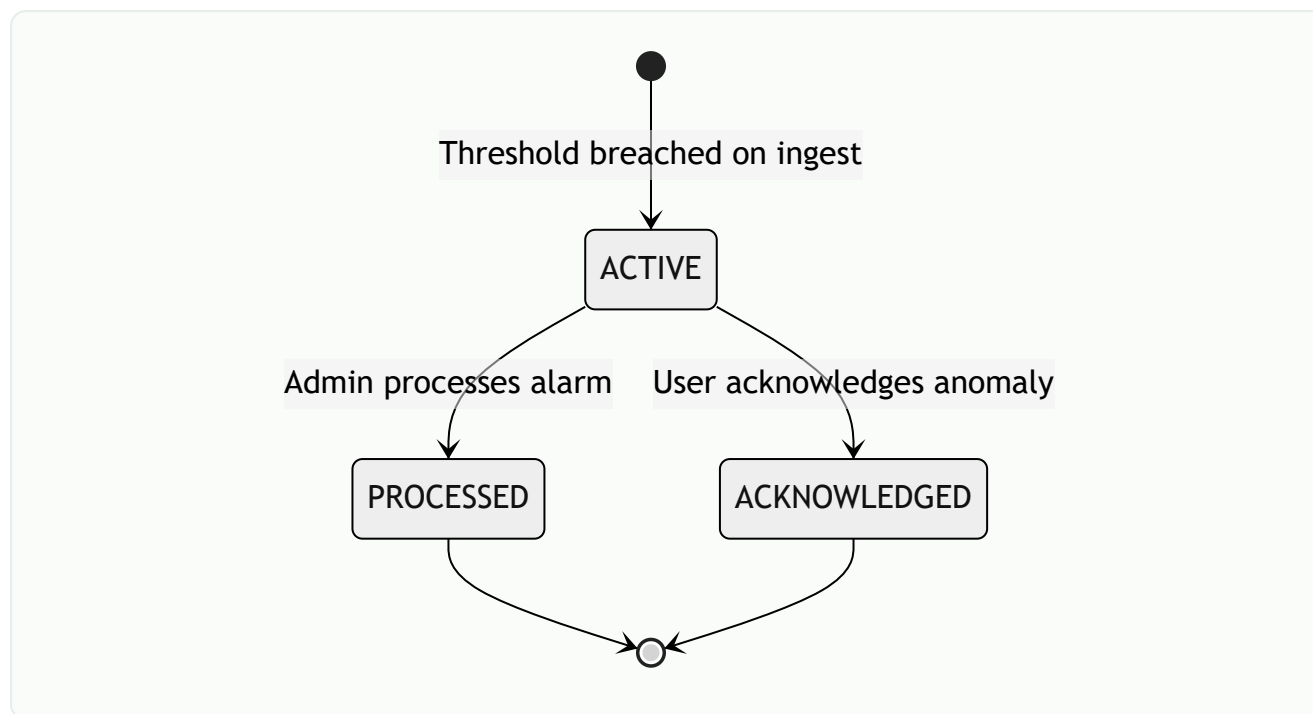
Variable	Unit	Source
SoilMoisture	%	MQTT M or simulator
BatteryLevel	%	MQTT B
TxCounter	count	MQTT TX

Example — **Electrical panel:**

Variable	Unit
VoltageA/B/C	V
CurrentA/B/C	A
PowerConsumption	kWh
PowerFactor	—

UI shows **actual configured variables**, not a fixed hardcoded list.

Notification & alarm lifecycle



Channels: in-app `Notification` , email queue, Socket `alarm:new` .

Subscription & products (commercial)

- Public product catalog (`GET /api/products`).
- Users submit subscription requests (`POST /api/subscriptions`).
- Super admin approves status (`PATCH /api/subscriptions/:id/status`).
- User dashboard shows current plan tier.

Feature availability matrix

Feature	Super Admin	Org Admin	User
All organizations	✓	—	—
Org devices CRUD	✓	✓	—
View assigned devices	✓	✓	✓
Ingest (field)	✓	✓	✓
Remote switch	✓	✓	—
Platform settings	✓	—	—

Feature	Super Admin	Org Admin	User
Themes / icons	✓	—	—
AI analytics pages	—	—	✓
Slab rates	—	—	✓
Alarm process (admin API)	✓	✓	limited

Related documents

- [System flows](#)
- [Web frontend](#)
- [Backend](#)